# The Cheon–Hwang Sub-Dittert Conjecture at $k = 3$, for Every Dimension

with a decomposition of the deficit uniform in $n$ and $k$

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**Abstract.** Cheon and Hwang conjectured in 1992 that $E_k(r) + E_k(c) - P_k(A) \le 2 - k!/n^k$ for every non-negative $n \times n$ matrix of total sum $n$ and every $1 \le k \le n$, with equality only at $J_n/n$; the endpoint $k = n$ is Dittert's conjecture. We prove the case $k = 3$ for every $n \ge 4$ — the inequality, the equality case, and a quantitative strengthening of both — by a single Positivstellensatz certificate whose nineteen symmetry-reduced coefficients are explicit rational functions of $n$. The strengthening is

$$(2 - 6/n^3) - [E_3(r) + E_3(c) - P_3(A)] \ge \theta_2(n) \cdot \|A - J_n/n\|_F^2$$

with $\theta_2(n) = (n^4 + 40n^2 - 84n + 40)/(n^5(n-1)^3(n-2))$, which contains the equality case as the special case where the left side vanishes. Together with Hwang's 1987 theorem at $n = 3$ this settles $k = 3$ in every dimension. The proof is uniform in $n$ because the governing degree is $k$, not $n$: the symmetry-reduced program has the same $12 \times 19$ shape at every dimension, so its solution is computed once over $\mathbb{Q}(n)$. Positive definiteness of the two $n^2 \times n^2$ Gram matrices is reduced in closed form — Bose–Mesner algebra of the rook's graph for one, the representation theory of $S_{n-1}$ for the other — to the positivity of ten rational functions of $n$ (nine independent; the tenth equals one of the other nine divided by $n$), each decided on $n \ge 4$ by a Sturm sequence over $\mathbb{Q}$.

The $k = 3$ certificate sits inside an exact decomposition of the deficit that is uniform in $n$ and in $k$. The case $k = 2$, which is classical, follows from that decomposition in a few lines; we give the derivation and a machine-checked proof. We also formalise, for every $2 \le k \le n$, the confinement of any violator to a shrinking neighbourhood of the Birkhoff polytope — at $k = n$ this is Theorem 2.1 of Cheon and Wanless 2012, and the extension to all $k$ is a routine transfer of their method — together with the Newton and Maclaurin inequalities that the confinement consumes, neither of which is in Mathlib. Every result stated below as a theorem is proved in Lean 4 with no `sorry` and on Lean's standard axioms, and is cited by its declaration name.

## 1 Introduction

Let

$$K_n = \left\{ A \in \mathbb{R}^{n \times n} : A_{ij} \ge 0, \sum_{i,j} A_{ij} = n \right\}, \tag{1}$$

and for $A \in K_n$ let $r$ and $c$ be its vectors of row and column sums. Write $\sigma_k(A)$ for the sum of the permanents of all $k \times k$ submatrices of $A$ — rows chosen from one $k$-subset, columns from another, independently — and put

$$E_k(v) = \frac{e_k(v)}{\binom{n}{k}}, \quad P_k(A) = \frac{\sigma_k(A)}{\binom{n}{k}^2}, \quad \gamma(n,k) = \frac{k!}{n^k}, \tag{2}$$

with $e_k$ the elementary symmetric function. Write $\Phi_k(A) = E_k(r) + E_k(c) - P_k(A)$.

**The conjecture (Cheon and Hwang [1], 1992).** For every $A \in K_n$ and every $1 \le k \le n$,

$$\Phi_k(A) \le 2 - \gamma(n,k), \tag{3}$$

with equality only at $A = J_n/n$.

The endpoints are familiar. At $k = n$ one has $\sigma_n = \mathrm{per}$ and $E_n(v) = \prod_i v_i$, so the statement is $\prod_i r_i + \prod_j c_j - \mathrm{per}(A) \le 2 - n!/n^n$: this is **Dittert's conjecture** verbatim, Conjecture 28 of the Cheon–Wanless survey [5]. At $k = 1$ the inequality is an identity **on $K_n$** — not on all of $\mathbb{R}^{n\times n}$ — since the difference is a constant multiple of $\sum_{ij} A_{ij} - n$; we note the distinction because a test that treats it as a formal identity fails. The cases $k \le 2$ for every $n$, and every $k$ at $n \le 3$, are known.

The interesting range is therefore $2 < k < n$, and it is that range which appears to have gone untouched. This paper settles the whole line $k = 3$.

## 1.1 Results

**Theorem A ($k = 3$, every dimension).** For every $n \ge 4$ and every $A \in K_n$,

$$E_3(r) + E_3(c) - P_3(A) \le 2 - \frac{6}{n^3}, \tag{4}$$

with equality if and only if $A = J_n/n$. More precisely, writing $\|\cdot\|_F$ for the Frobenius norm,

$$\begin{gathered}\left(2 - \frac{6}{n^3}\right) - [E_3(r) + E_3(c) - P_3(A)] \ge \theta_2(n)\cdot \|A - J_n/n\|_F^2,\\ \text{where}\quad \theta_2(n) = \frac{n^4 + 40n^2 - 84n + 40}{n^5(n-1)^3(n-2)},\end{gathered} \tag{5}$$

and $\theta_2(n) > 0$ for every $n \ge 4$.

Lean: `SubDittertK3.subDittert_k3_full`.

The three statements are one statement. Equation 5 implies Equation 4 because $\theta_2(n) > 0$, and it implies the equality case because $\|A - J_n/n\|_F = 0$ only at $J_n/n$; conversely $\Phi_3(J_n/n) = 2 - 6/n^3$ exactly. We state all three because the quantitative form is what the certificate actually produces, and it is stronger than what the conjecture asks for: the functional falls away from its maximum at least quadratically in the distance to the maximiser, at an explicit rate. The constant $\theta_2(n)$ is not claimed optimal; see §9.4.

**Corollary B (two layers, stated as such).** With Hwang's theorem for Dittert at $n = 3$ [3], the case $k = 3$ of the Cheon–Hwang conjecture holds, with its equality case, for every $n \ge 3$. Since $k = 3$ requires $n \ge 3$, no dimension is left open on this line.

**This corollary is not a single-layer result and should not be cited as one.** The range $n \ge 4$ is Theorem A and is machine-checked (`SubDittertK3.subDittert_k3_full`). The remaining case $n = 3$ is Hwang's refereed theorem of 1987 [3], which we do not reprove in Lean. The corollary's support is therefore Lean **plus** the published literature, and it inherits whatever reliance one places on [3].

Since $2 < 3 < n$ for every $n \ge 4$, Theorem A lies inside the intermediate range. On the evidence of §2 it is the first resolved case of the Cheon–Hwang conjecture with $2 < k < n$. That is a priority claim rather than a claim of difficulty, and §2 records why it is perishable.

The certificate is a construction special to $k = 3$, but it sits inside a decomposition that is not.

**Theorem C (the deficit, decomposed uniformly in $n$ and $k$).** Write $A = J_n/n + b$, let $R$ and $C$ be the row and column sums of $b$, and put

$$s_d = \frac{[k]_d}{[n]_d}, \quad t_d = s_d^2 \cdot \frac{(k-d)!}{n^{k-d}} \tag{6}$$

with $[x]_d$ the falling factorial. Then for every $1 \le k \le n$, identically in $b$,

$$(2 - \gamma(n,k)) - \Phi_k(A) = \sum_{d=1}^{k} [t_d \cdot \sigma_d(b) - s_d \cdot (e_d(R) + e_d(C))]. \tag{7}$$

Lean: `SubDittertUniversal.universal_identity`.

Equation 7 is elementary, and no novelty is claimed for it: it is the $\sigma_k$ analogue of the classical expansion $\operatorname{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\,\sigma_j(A)$, combined with $e_k(\mathbf{1} + R) = \sum_d \binom{n-d}{k-d} e_d(R)$. Its role here is that it makes every $k$-dependence explicit and finite, so that statements which look like separate facts at separate $k$ become instances of one identity. The absence of a $d = 0$ term is the reason the constant $2 - \gamma(n,k)$ is exactly right, at every $n$ and $k$ at once.

**Theorem D ($k = 2$, every dimension).** For every $n \ge 2$ and every $A \in K_n$,

$$\Phi_2(A) \le 2 - 2/n^2. \tag{8}$$

With $\kappa = 2/(n(n-1))$ the deficit is a manifest sum of squares on the hyperplane $\sum_{ij} b_{ij} = 0$:

$$(2 - 2/n^2) - \Phi_2(A) = \frac{1}{2}\left[\kappa^2\|b\|^2 + \kappa(1-\kappa)(\|R\|^2 + \|C\|^2)\right]. \tag{9}$$

Lean: `SubDittertK2.subDittert_k2`. **The statement is classical** (§2.2); the contributions are the derivation from Equation 7 and the machine-checked proof.

Theorem D needs no Cauchy–Schwarz, no Gram matrix and no Maclaurin inequality: Equation 9 falls out of Equation 7 at $k = 2$, and both coefficients are non-negative for $n \ge 2$ because $\kappa \le 1$ there. Its interest is as a control on the uniform machinery — it is the one case where that machinery can be run end to end against an answer that is independently known — and, with Theorem A, it makes the band $k \in \{2, 3\}$ complete.

Two further results are recalled rather than claimed. Both are formalised here, and it is the formalisation, not the mathematics, that is ours.

**Theorem E (confinement; Cheon and Wanless [6], Theorem 2.1, at $k = n$; transferred to all $k$).** Let $2 \le k \le n$ and $A \in K_n$. Then

$$\frac{\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2}{n(n-1)} - \gamma(n,k) \le (2 - \gamma(n,k)) - \Phi_k(A), \tag{10}$$

so any $A$ violating the Cheon–Hwang bound satisfies

$$\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2 \le \frac{(n-1)k!}{n^{k-1}}. \tag{11}$$

Lean: `SubDittertMaclaurin.theoremM'`, `SubDittertMaclaurin.confinement'`.

At $k = n$ this is Theorem 2.1 of [6] in contrapositive form, with subset sums in place of the $\ell^2$ norm and a threshold of the same scale; that paper describes its own result as significantly extending a result of Hwang, so the idea is older still. Pang [9] runs the same argument at $k = n$ and refines the quantitative step without changing its shape; that item is an unrefereed preprint. The only part not found in the literature we checked is the transfer from the Dittert functional to the whole Cheon–Hwang family uniformly in $k$, **and that is a routine transfer of a known method**. It is formalised here for two reasons that have nothing to do with novelty: it is elementary enough to check, and it is uniform in $k$, so one Lean proof covers the whole family.

**Theorem F (Newton; Maclaurin).** For every real-rooted real polynomial and every admissible index, Newton's inequality holds; in vector form, for $v \in \mathbb{R}^n$ and $1 \le j \le n-1$,

$$e_{j-1}(v)\, e_{j+1}(v) \binom{n}{j}^2 \le e_j(v)^2 \binom{n}{j-1}\binom{n}{j+1}, \tag{12}$$

with no positivity hypothesis. Consequently, for $v \ge 0$ with $\sum_i v_i = n$ and $2 \le k \le n$, $E_k(v) \le E_2(v)$.

Lean: `NewtonIneq.newtonAt_all`, `NewtonIneq.newton_esymmF`, `NewtonIneq.pnorm_le_two`. **Classical mathematics**; the contribution is the formalisation.

The telescoped form $E_k \le E_2$ carries no fractional powers, unlike the usual $E_k^{1/k} \le E_2^{1/2}$; that is what makes it the statement to formalise, and it is exactly what Theorem E consumes. Neither Newton's inequalities nor Maclaurin's inequality is in Mathlib v4.14.0, whose `NewtonIdentities` file carries Newton's **identities** only.

### 1.2 Support layers

Every result above is stated with the layer it rests on, because in this corner of the literature the distinction has recently been expensive.

- **Theorems A and C–F are proved in Lean 4**, on Lean's standard axioms, with no `sorry` anywhere in the development and no use of `native_decide`. §9 states the verification standard precisely and pins it to a commit. Theorem A is Lean-proved in all three parts, equality case and stability bound included.
- **Corollary B** rests on Theorem A for $n \ge 4$ and on the published literature at $n = 3$.
- **The fixed-dimension certificates of §7.1 are not Lean-checked.** They are exact rational data accepted by an independent standalone verifier. They are recorded as anchors, and no theorem above depends on them.
- **§10 is computational evidence, not a theorem of this paper**, and is not Lean-checked. Its verdicts are exact rational computations with stored witnesses.
- **Nothing here is refereed**, including this paper. Exact verification and machine checking are different things from refereeing, and weaker ones.

## 2 The state of the line, and of the neighbouring endpoint

### 2.1 The $k = n$ endpoint moved in one week, publicly and unrefereed

Dittert's conjecture is the endpoint of the same family, and its status changed sharply in July 2026. Only two dimensions rest on refereed work: $n = 2$ (Sinkhorn [4]) and $n = 3$ (Hwang [3]). Everything else that is currently claimed is either an unrefereed preprint or a public code repository:

| dimensions | claimed by | status |
|---|---|---|
| $n = 2$ | Sinkhorn [4], 1984 | refereed |
| $n = 3$ | Hwang [3], 1987 | refereed |
| $n = 4$ | Divya K. U. and Somasundaram, arXiv:2312.00464 | **claim withdrawn in the published version** [8] |
| $n = 4$ | public repository [11], 21 Jul 2026 | public, dated, unrefereed |
| $n = 4, 5, 8$–$16$ | public repository [11], 23 Jul 2026 | public, dated, unrefereed |
| $n = 6$–$15$ | public repositories [11], 24 Jul 2026 | public, dated, unrefereed |
| $n = 16$ | Kafidov [10], arXiv:2607.19439, 21 Jul 2026 | preprint, no DOI |
| $n \geq 17$ | Pang [9], arXiv:2606.01531, 1 Jun 2026 | preprint, no DOI |

Table 1: Public dated claims on Dittert's conjecture ($k = n$), as of 28 July 2026. Refereed status is stated as it is, not as one would like it.

Two features of the table bear on how the present paper should be read. At the 28 July 2026 sweep none of the repositories listed [11] had a corresponding preprint or journal article, so the indexed literature does not on its own record the state of this conjecture. And the record moves in both directions: the v1 abstract of arXiv:2312.00464 claims Dittert at $n = 4$, while the published version in *Special Matrices* **12** (2024) [8] contains no such claim; the preprint record still shows v1.

None of the unrefereed items is asserted here to be wrong. They are public and dated, which is what matters for priority, and none of them is refereed, which is what matters for reliance.

## 2.2 The intermediate range $2 < k < n$

The generalisation itself has attracted almost nothing. The Cheon–Hwang paper [1] has five citing papers — Cheon and Wanless 2012 [6], Cheon and Wanless 2007 [7], the Cheon–Wanless survey [5], and Cheon and Yoon 2006 and Cheon 1993 (both [12]) — and none of them works on the generalisation. Reference [6] says only that Cheon and Hwang "posed a problem generalising the Dittert conjecture"; its own results concern partly decomposable matrices and asymptotics. To our knowledge neither the literature nor any public code repository carries a result on the intermediate range; the closest item is the one described next.

The one adjacent effort is a `subdittert/` package inside one of the repositories of [11]. Its README is explicit about scope, and we quote it verbatim:

> "The endpoint `k=n` is Dittert's problem. The cases `k<=2` are historically known. This package does **not** claim the unresolved intermediate cases."

That quotation is also the clearest available statement of the prior status of Theorem D: $k \leq 2$ is historically known, for every $n$, and we claim no novelty for the statement. Its results are $k = n - 1$ for $18 \leq n \leq 80$, $k = n - 2$ for $18 \leq n \leq 40$, and a table over 425 pairs with $17 \leq n \leq 40$ marked conditional on a scaling hypothesis. Note that $(n, k) = (4, 3)$ **is** a $k = n - 1$ case, so that package works the same diagonal — but from $n = 18$ upward, by a large-$n$ scaling argument that cannot reach $n = 4$, and in part conditionally. It does not intersect the line $k = 3$ except at $n = 4$, which it excludes.

One item is flagged rather than paraphrased: Cheon and Wanless 2007 [7] is paywalled and was not obtained, and is reported to settle $n \leq 3$ for all $k$. Nothing in this paper depends on it: Corollary B takes $n = 3$ from Hwang [3], and §7.1 reproves it by certificate.

## 3 The decomposition, and the case $k = 2$

Theorem C is proved by reading the two definitions and dividing. Since $A = J_n/n + b$ has row sums $\mathbf{1} + R$, $e_k(\mathbf{1} + R) = \sum_d \binom{n-d}{k-d} e_d(R)$; and since a term of a permanent is a bijection, $\sigma_k(J_n/n + b) = \sum_d \binom{n-d}{k-d}^2 (k-d)!\, n^{-(k-d)} \sigma_d(b)$. Dividing by $\binom{n}{k}$ and $\binom{n}{k}^2$ turns $\binom{n-d}{k-d}/\binom{n}{k}$ into $[k]_d/[n]_d = s_d$. The $d = 0$ terms are 1, 1 and $\gamma(n, k)$, which cancel the constant $2 - \gamma(n, k)$ exactly; that cancellation is why the sum in Equation 7 starts at $d = 1$.

Three consequences are worth naming, because each has been observed separately in this family and each is the same identity read at one degree. At $d = 1$ the gradient of the deficit at $J_n/n$ is $\left(-2k/n + k \cdot k!/n^{k+1}\right)\mathbf{1}$, so $J_n/n$ is critical for every $(n, k)$. At $d = 2$ the Hessian has exactly two distinct eigenvalue parameters, $s_2 = k(k-1)/(n(n-1))$ and $t_2 = s_2^2 (k-2)!/n^{k-2}$, again for every $(n, k)$. And the coefficient of a degree-$d$ monomial of the deficit takes one of three values — $-2s_d + t_d$ if the cells form a partial permutation, $-s_d$ if they have distinct rows or distinct columns but not both, and 0 otherwise — so three numbers per degree describe the whole objective, whatever $k$ is. The closed forms used in §6.1 are the case $k = 3$ of that rule.

At $k = 2$ the sum has two terms and both are computable in closed form. Using $2\sigma_2(b) = \left(\sum b\right)^2 - \|R\|^2 - \|C\|^2 + \|b\|^2$ and $2e_2(R) = \left(\sum R\right)^2 - \|R\|^2$, and restricting to $\sum b = 0$, the deficit collapses to Equation 9, which is Theorem D. The identity was checked against the objective built from the 1992 definition in exact rational arithmetic at $n = 2, ..., 6$, four random $b$ per $n$ on the hyperplane, with no mismatches; the Lean proof of `subDittert_k2` goes through Equation 7 and does not use that check.

The same route does not reach $k = 3$. There the deficit is not a sum of squares on the hyperplane, and the certificate of §5 is needed.

## 4 The objective at $k = 3$, and how we know it is the right one

A certificate for the wrong polynomial is worthless and looks identical to a certificate for the right one, so the construction of the objective is checked before anything else. In centred coordinates $b = A - J_n/n$ put

$$F(b) = (2 - \gamma(n, k)) - [E_k(r) + E_k(c) - P_k(A)], \tag{13}$$

a polynomial of degree $k$ over $\mathbb{Q}$; the conjecture at $(n, k)$ is $F \geq 0$ on $K_n$. Five independent tests are run on the construction, all passing:

1. **$\boldsymbol{\sigma_k}$ by two structurally different algorithms.** One enumerates all $\binom{n}{k}^2$ pairs of index sets and sums $k!$ products. The other never forms a submatrix at all: it uses $\mathrm{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\,\sigma_j(A)$ in $\mathbb{Q}[x]$, with the order-$n$ permanent by Ryser inclusion–exclusion over column subsets. Forty random rational matrices agree, for $2 \leq n \leq 5$ and every $k$.
2. **Symbolic against from-scratch** evaluation at random rational points, at $(3, 2)$, $(4, 3)$, $(4, 4)$ and $(5, 3)$.
3. **A $\boldsymbol{k = n}$ positive control.** The polynomial built here at $(4, 4)$ is compared coefficient by coefficient with the one from an independent Dittert pipeline: 1040 monomials each — that is the number carrying a non-zero coefficient in the degree-4 objective, out of the 4845 of degree at most 4 in the sixteen entries — and zero differing coefficients. That pipeline already underlies a verified certificate, so it is ground truth rather than a mirror.
4. The numerical trap of §4.1, explicitly separated.
5. $k = 1$ is confirmed to be an identity on $K_n$, and equality at $J_n/n$ is confirmed for every tested $(n, k)$.

### 4.1 Two coincidences that make the bound useless as a check

$2-\gamma(4,3)=2-6/64$ and $2-\gamma(4,4)=2-24/256$ are **both** $61/32$. The same collision recurs at $n=5$: $2-\gamma(5,4)=2-\gamma(5,5)=1226/625$, which is the published Dittert value at $n=5$. So seeing the right constant is **no evidence whatsoever** that the $k=3$ or $k=4$ code is doing what it claims.

The two cases are separated by comparing whole polynomials rather than constants. At $n=4$ the $k=3$ and $k=4$ objectives have degrees 3 and 4; their supports together carry 1040 monomials — the $k=3$ support, 552 monomials, sits inside the $k=4$ one — and the two disagree at every one of the 1040. At $n=5$ the same comparison spans 14005 monomials, degrees 4 and 5, again disagreeing at all of them.

# 5 The certificate

## 5.1 Form

In centred coordinates the ansatz is

$$F(b)=\sigma_0(b)+\sum_p\left(\frac{1}{n}+b_p\right)\sigma_p(b)+\lambda(b)\cdot\left(\sum_q b_q\right), \tag{14}$$

where $\sigma_0$ and every $\sigma_p$ is a sum of squares and $\lambda$ is a free polynomial. On $K_n$ the last term vanishes, since $\sum_q b_q=0$ there, and every $1/n+b_p=A_p\geq 0$; so the right side is a sum of non-negative terms and $F\geq 0$ on $K_n$, which is the theorem.

The shape is Putinar's [2]: one sum-of-squares multiplier for each defining inequality $A_p\geq 0$, a free multiplier for the equality $\sum_q b_q=0$, truncated at a fixed degree. No existence theorem is invoked. Putinar's applies to polynomials **strictly** positive on the set, and $F$ vanishes at $J_n/n$, in the relative interior of $K_n$; the representation is exhibited, not deduced, and the work is in getting it at the degree $F$ already has.

**Why the degree closes.** Here $\deg F=k$, not $n$. At $k=3$ a Gram basis of degree 1 gives $\deg\sigma\leq 2$ and $\deg\big(\sigma_p b_p\big)\leq 3$, matching $\deg F$ exactly with no surplus top band to cancel. The Gram matrices are $n^2\times n^2$. This is the structural reason the whole line $k=3$ is accessible while a fixed $k=n$ is not: in the Dittert case the objective's degree grows with the dimension and the ansatz cannot follow it.

**Why rounding works.** The standing obstruction to exact rational rounding is that a tight bound forces a singular optimal Gram. Here the Hessian of $F$ at $J_n/n$ restricted to the tangent space $\{\sum X=0\}$ is positive definite — at $(4,3)$ its characteristic polynomial on the projected space is exactly $x(x-1/16)^9(x-29/16)^6$ over $\mathbb{Q}$, and the multiplicities $9=(n-1)^2$ and $6=2(n-1)$ are forced by Schur's lemma, since the tangent space is $(V|1)\oplus(1|V)\oplus(V|V)$ with transposition fusing the first two. So there is no forced kernel inside the tangent space: centring at the extremiser and excluding the constant monomial removes the only degenerate direction.

## 5.2 Symmetry reduction, and the fact that its shape is fixed in $n$

The problem is invariant under $(S_n\times S_n):\mathbb{Z}_2$ acting by row permutations, column permutations and transposition. Reducing Equation 14 by that group leaves, **at every $n$ alike**:

| object | count, independent of $n$ |
|---|---|
| orbit constraint rows | $1+1+3+7=12$, of rank 11 over $\mathbb{Q}(n)$ |
| $\sigma_0$ variables | 3 |
| $\sigma_{11}$ variables | 11 |
| $\lambda$ variables | 5 |
| cone size | $n^2$ (the only thing that grows) |

Table 2: The symmetry-reduced program at $k=3$ with a degree-1 Gram basis.

Nineteen unknowns and twelve equations, at $n=4$, $n=5$, $n=6$ and beyond. This is representation stability, and here — unlike in the $k=n$ case, where the same stability is undercut by the growing degree — there is no counteracting obstruction. That observation is what makes a single symbolic solve plausible; the rest of the paper is the work of turning it into a proof.

One implementation point matters for reaching large $n$ at all. The transporter carrying the corner orbit to position $(i,j)$ is the explicit product of the row transposition $(0\,i)$ with the column transposition $(0\,j)$, at cost $O(n^2)$; enumerating the group instead costs $2(n!)^2$, which is $5.1\times 10^7$ already at $n=7$. The reduced system is provably unchanged, because two transporters differ by an element of the stabiliser of $(0,0)$, which preserves each $\sigma_{11}$ orbit.

# 6 The uniform certificate

## 6.1 The constraint system in closed form, over $\mathbb{Q}(n)$

Both halves of the $12\times 19$ system are **derived** as functions of $n$, not interpolated; "no fit is used anywhere" is a stronger statement than "the fit was validated".

**The right-hand side.** The coefficient in $F$ of an arbitrary monomial is closed form for all $n$ at once. Writing $d$ for the degree of the monomial and reading the two definitions directly,

```
[monomial] e_k(r)     = C(n-d, k-d)   if the d cells lie in distinct ROWS, else 0
[monomial] sigma_k    = C(n-d, k-d)^2 (k-d)! n^{-(k-d)}
                                      if the cells form a PARTIAL PERMUTATION, else 0
```

because within one product $r_i r_j r_k$ no index repeats, so two cells in a row cannot occur; and a term of a permanent is a bijection, so the cells must have distinct rows **and** distinct columns. This is the case $k=3$ of the coefficient rule of §3. Checked against the fully expanded objective at $n=4,5,6$: 969, 3276 and 9139 monomials, no mismatches. Those counts are of **every** monomial of degree at most 3 in the $n^2$ entries, absent ones included — the closed form has to return zero exactly where the expanded objective has no term, so the check is over the whole coefficient vector and not over the support. As a sanity check it gives the degree-one coefficient as $-6/n+18/n^4$, which at $n=4$ is $-183/128$, the gradient value computed independently.

**The orbit sizes.** By orbit–stabiliser, $|\text{orbit}(P)| = 2\cdot[n]_r\cdot[n]_c/\ |\text{Stab } P|$ with falling factorials, and $|\text{Stab } P|$ found by brute force over at most $2\cdot 3!\cdot 3! = 72$ relabellings. A multiset of at most three cells meets at most three rows and three columns, so every orbit size is a polynomial in $n$ of degree at most 6. Checked against brute-force enumeration at $n=4,5,6,7$, including the stabiliser case: no mismatches.

The structure that makes this work is that with a degree-1 basis the product $\text{basis}[u]\cdot\text{basis}[v]$ **is** the pair $\{u,v\}$, so the $\sigma_0$ variable orbits and the constraint-row orbits are the same objects; and canonicalisation is equivariant, so both incidence blocks are an orbit size times a 0/1 incidence.

**Cross-check against the trusted path.** The symbolic system was compared entry by entry, right-hand side included, with the system built by the original code path — the one that produced the already-verified fixed-$n$ certificates — at $n = 4, 5, 6$. No mismatches. The two share no logic: one does union–find over every monomial of the expanded objective, the other evaluates closed forms.

Row-reducing over the field $\mathbb{Q}(n)$ gives: 12 equations, 19 unknowns, rank 11, one dependent row, **consistent**, with an 8-dimensional affine family of solutions. So the linear half of the certificate exists at every $n$ at once.

## 6.2 Definiteness reduces to ten rational functions of $n$

What remains is to choose within that family so that both Gram matrices are positive definite for every $n \geq 4$. Both are block-diagonalised **in closed form** — a numerical basis at one $n$ could not settle all $n$.

$\boldsymbol{\sigma_0}$**.** Its Gram is $aI + bA_1 + cA_2$ in the Bose–Mesner algebra of the rook's graph $K_n \square K_n$, so it has exactly three eigenvalues:

$$\begin{aligned} \theta_0 &= a + 2(n-1)b + (n-1)^2 c && \text{multiplicity } 1, \\ \theta_1 &= a + (n-2)b - (n-1)c && \text{multiplicity } 2(n-1), \\ \theta_2 &= a - 2b + c && \text{multiplicity } (n-1)^2. \end{aligned} \tag{15}$$

$\boldsymbol{\sigma_{11}}$**.** With $V'$ the standard $(n-2)$-dimensional representation of $S_{n-1}$ one has $\mathbb{R}^{n^2} = 4(1|1) \oplus 2(V'|1) \oplus 2(1|V') \oplus (V'|V')$, and transposition fuses $(V'|1)$ with $(1|V')$ and splits the four-dimensional trivial multiplicity space as $3+1$. The blocks are a $3 \times 3$ (call it $A$), a $1 \times 1$ sign block $B$, a $2 \times 2$ block $C$ of multiplicity $2(n-2)$, and a $1 \times 1$ block $D$ of multiplicity $(n-2)^2$.

Both consistency checks pass: $\sum d(d+1)/2 = 6 + 1 + 3 + 1 = 11$, the number of orbit parameters, and $3 + 1 + 2 \cdot 2(n-2) + (n-2)^2 = n^2$. The decomposition was verified against a direct eigendecomposition of the assembled $n^2 \times n^2$ matrix on **random** orbit coefficients at $n = 4, \ldots, 8$ — full spectrum against the union of block spectra with predicted multiplicities, worst discrepancy $1.4 \times 10^{-13}$. Random coefficients matter here: a check run on the real certificate could pass by accident on a matrix carrying extra structure.

**Consequence.** Positive definiteness of both Grams, for all $n \geq 4$, is exactly the positivity of **ten** explicit rational functions of $n$: the three $\sigma_0$ eigenvalues, the three leading principal minors of $A$, the block $B$, the two leading minors of $C$, and the block $D$.

One of these ten is not independent of the rest: the block $D$ equals $\theta_2/n$ **exactly**, an identity of rational functions rather than a coincidence at sampled $n$ — the two share the same numerator and differ only by a factor of $n$ in the denominator. So positivity of $D$ follows immediately from positivity of $\theta_2$ together with $n > 0$, and the certificate rests on **nine** independent positivity facts, not ten; $D$ is carried along as a consequence rather than an additional condition. We report and certify all ten regardless — Sylvester's criterion asks for all ten directly, and the Sturm and Lean machinery below decides each on the same uniform footing — but a reader should not credit the certificate with ten independent facts where nine suffice.

These ten were validated first on the independently produced, already-verified certificates at $n = 4, 5, 6$: all ten strictly positive in each case.

## 6.3 The geometry of the feasible set

Choosing the eight free variables as functions of $n$ is the whole difficulty, and the shape of the difficulty is not the obvious one.

**The feasible set is unbounded, and the recession cone is exactly two-dimensional.** It is derived, not probed. A recession direction is a homogeneous solution with the sigmas still positive semidefinite; on the hyperplane $\sum b = 0$ every term of Equation 14 is non-negative on $K_n$, so each vanishes there, and a sum-of-squares quadratic form vanishing on a hyperplane has Gram $\mathbf{1}u^T + u\mathbf{1}^T$, which is positive semidefinite only for $u$ a non-negative multiple of $\mathbf{1}$. Hence the recession cone adds $c_0 J$ to the $\sigma_0$ Gram and $c_1 J$ to the $\sigma_{11}$ Gram, $c_i \geq 0$, with $\lambda$ determined. This was confirmed as an exact kernel element over $\mathbb{Q}(n)$ with a round trip through the free coordinates.

**The lineality space is larger, and it is what makes optima drift.** The quantities that actually have to be controlled need $G$ and $H$ positive semidefinite only on $\mathbf{1}^\perp$, which gives $G = \mathbf{1}u^T + u\mathbf{1}^T$ and $H = \mathbf{1}v^T + v\mathbf{1}^T$ with $u$ forced to a multiple of $\mathbf{1}$ by transitivity but $v$ free on the three stabiliser classes. Its four generators have an invertible $4 \times 4$ minor on the four free $\lambda$ columns, so the lineality space is exactly the $\lambda$ reparametrisation, and $\{f_{15} = f_{16} = f_{17} = f_{18} = 0\}$ is a transversal slice. On that slice the recession cone is trivial and **the feasible set is compact**. The essential design problem has **four** unknowns. Numerically, drift along the lineality space is plainly visible: the scaled free variable $f_6 n^3$ takes the values $149, 20.3, 9.69, 35311, 13.3, 83681, 6.66$ at $n = 20, 50, 100, 200, 500, 1000, 2000$, which no curve in $n$ will follow.

**The essential set is a sliver, and that is why no fit can hold it.** In the scaled coordinates $\beta = f \cdot n^3$, four of the ten quantities are differences of terms of size $n^2$ or $n^3$ that must cancel down to $O(1)$. Working the cancellations out forces

$$\beta_6 - 2\beta_9 = O(n^{-2}), \quad \beta_{12} - (2\beta_9 - 1) = O(n^{-1}), \quad \beta_{11} - 2\beta_{12} - 2 = O(n^{-1}), \tag{16}$$

the last pinned to $O(n^{-3})$; asymptotically $\beta \to (2b, b, 4b, 2b-1)$, a one-parameter family with $1/2 < b < 1$, with the analytic centre at $b \approx 0.8486$. A least-squares fit whose residual is small against the diameter of the essential set is still far outside it, and a relation pinned to $O(n^{-3})$ is beyond the reach of curve fitting at any degree.

## 6.4 Adapted coordinates, and one exact elimination

Write the cancellations into the coordinates, so that they happen symbolically rather than numerically:

$$\beta_9 = b, \quad \beta_6 = 2b + \frac{x}{n^2}, \quad \beta_{12} = 2b - 1 + \frac{y}{n}, \quad \beta_{11} = 2\beta_{12} + 2 + \frac{z}{n}. \tag{17}$$

Then **solve for $z$ exactly over $\mathbb{Q}(n)$ from the equation $\theta_2 = D$**, rather than fitting it. This is the step that makes the difference. The two quantities are affine in $z$ with opposite-sign coefficients of size $n^2$, and the window between them has width $O(n^{-2})$; at $n = 10^6$ a numerical fit would need twelve correct digits merely to stay inside. With $z$ eliminated, the remaining parameters carry $\Theta(1)$ coefficients and an ordinary semidefinite program over a grid of $n$ finds them. The answer is simple:

$$b = 1, \quad x = 8 + \frac{20}{n}, \quad y = -2 - \frac{10}{n}, \tag{18}$$

$$z = \frac{6n^7 - 28n^6 + 41n^5 - 28n^4 + 48n^3 - 164n^2 + 208n - 80}{n^7 - 7n^6 + 19n^5 - 25n^4 + 16n^3 - 4n^2}. \tag{19}$$

The window $1/2 < b < 1$ of §6.3 and the choice $b = 1$ here are not in conflict, and the gap between them is worth naming. That window constrains the **limit** of the scaled quantities; the certificate is not the limit point. Its finite-$n$ corrections — $x/n^2$ and $y/n$ above, and $z$ solved exactly — are what carry the four tight quantities, and they are non-zero at every finite $n$. What the endpoint costs is margin, not sign: $\theta_2$ decays like $n^{-5}$. Positivity is then decided at every $n \geq 4$ by the Sturm step of §6.6, which acts on the exact rational functions and is indifferent to how small the margin becomes.

The transferable form of this step is short. In a family of certificates indexed by a parameter, identify the tight direction and eliminate it symbolically; fit only the slack ones. Quotienting the lineality space out is necessary — without it the optima drift and nothing converges — but it is not sufficient, because the compact essential set that remains can still be a sliver.

Two of the ten quantities cost nothing, because they are pure gauge. Moving along the lineality space adds $\mu n^2$ to $\theta_0$ and $sw^T + ws^T$ to the block $A$, with $s = (1, 2(n-1), (n-1)^2)$ and $w$ free in $\mathbb{R}^3$; choosing $w$ so that $A$ becomes $\mathrm{diag}(\gamma, T^T AT)$ in a basis $(e, T)$ with $e = (1, 0, 0)$, and taking $\gamma = 1/n^3$ and $\theta_0 = 1/n$, reduces $A \succ 0$ to $T^T AT \succ 0$, which is already one of the essential conditions. No new positivity requirement appears.

The resulting nineteen certificate variables are exact rational functions of $n$; they are recorded in full in the accompanying material and are not reproduced here. Three of them are $1/n^3$ exactly; the largest has numerator of degree 9 and denominator of degree 12.

## 6.5 Obstructions to simpler designs

Four routes that a reader would otherwise try are closed, each by an exact rational witness.

- **Linear programming for the design step is infeasible**, at ansatz degrees $D = 0, 1, 2, 3, 4$ (8 to 40 unknowns, 350–458 coefficient rows). The blocker is not the ansatz: plain diagonal dominance **fails on the already-verified certificates** at $n = 4, 5, 6$, in row 0 of both the $3 \times 3$ and the $2 \times 2$ block, so no LP built on dominance can be feasible. The cause is scaling — the isotypic vectors are unnormalised, so diagonal entries differ by orders of magnitude — and a rescaling does exist (both blocks are H-matrices at the verified certificates), but the weights that work are read off the certificate that the LP is trying to choose. The design step is **bilinear** in weights and certificate, not linear.
- **Least squares through analytic centres**, at degrees 1 to 7, fails off-grid at $n = 17$, $n = 71$ and $n = 811$ — always between grid points, always on a tight quantity.
- **Pinning four entries to constant targets** fails for all 210 four-subsets. The centre's own entries move by a factor of four between $n = 4$ and $n = \infty$, so constant targets pull the solution out of the set at one end or the other. Separately, the subset $\{\theta_2, D, C_{01}, T_{01}\}$ is exactly singular over $\mathbb{Q}(n)$; 197 of the 210 are non-singular.
- **Restricting $\boldsymbol{\lambda}$ to be a sum of squares** is sound, since $\lambda$ multiplies an equality constraint, but it removes only the two-dimensional recession cone and not the four-dimensional lineality space that causes the drift.

## 6.6 Sturm certification of all ten quantities

Substituting $n = m + 4$ turns each quantity into a ratio of polynomials in $m$ with the range becoming $m \geq 0$. The sign of each denominator on that range is **checked**, not assumed. The numerator's positivity on $[0, \infty)$ is then decided — decided, not estimated — by a Sturm chain on the squarefree part, comparing sign variations at 0 and at $+\infty$. There is no sufficiency gap in this step, which is precisely why the tempting shortcut "all coefficients non-negative in $m$" is only a heuristic: $m^2 - m + 1$ is positive everywhere and has a negative coefficient.

| **quantity** | **sqfree deg** | $V(0)$ | $V(\infty)$ | **roots in** $(\mathbf{0}, \boldsymbol{\infty})$ |
|---|---|---|---|---|
| `G0.theta0` | constant 1 | — | — | 0 |
| `G0.theta1` | 6 | 1 | 1 | 0 |
| `G0.theta2` | 4 | 1 | 1 | 0 |
| `H.A minor1` | constant 1 | — | — | 0 |
| `H.A minor2` | 5 | 1 | 1 | 0 |
| `H.A minor3` | 13 | 3 | 3 | 0 |
| `H.B` | 5 | 2 | 2 | 0 |
| `H.C minor1` | 5 | 1 | 1 | 0 |
| `H.C minor2` | 12 | 2 | 2 | 0 |
| `H.D` | 4 | 1 | 1 | 0 |

Table 3: All ten quantities, positive for every $n \geq 4$. Each row is a complete decision, not a sample: no roots in $(0, \infty)$ and a positive value at $m = 0$. `H.D` and `G0.theta2` share the same numerator — visible already in the matching squarefree degree 4 — because $D = \theta_2/n$ exactly (§6.2); we certify both rows independently regardless.

All ten denominators are sign-definite on $n \geq 4$, as are the denominators of all nineteen certificate variables, so nothing blows up at any dimension.

# 7 Verification

A certificate is worth what its checking is worth. The independent verifier **re-derives the objective from the 1992 definition** — $e_k$ of the row and column sums, and $\sigma_k$ as an explicit double sum of subpermanents over pairs of $k$-subsets — and uses none of the closed forms of §6.1, none of the block decomposition of §6.2 and none of the Sturm machinery of §6.6. The only shared code is the orbit bookkeeping that inflates nineteen numbers into two $n^2 \times n^2$ matrices, and that is exercised against stored certificates and against deliberate mutations.

At each $n$ it checks three things: that both Gram matrices are symmetric and **positive definite**, by exact rational $LDL^T$ on the **full** $n^2 \times n^2$ matrices rather than through the blocks; that the identity Equation 14 holds at random rational points; and that the constant is $2 - k!/n^k$. Positive definiteness of the single Gram at the corner gives it for every multiplier, since each $\sigma_p$'s Gram is a permutation conjugate of it.

| $n$ | **Gram** | $\sigma_0$ $LDL^T$ | $\sigma_{11}$ $LDL^T$ | **identity** |
|---|---|---|---|---|
| $4, 5, 6, 7$ | 16–49 | PD | PD | exact over $\mathbb{Q}$, 2 points each |
| 12 | $144 \times 144$ | PD, $9.17 \times 10^{-6}$ | PD, $7.77 \times 10^{-7}$ | exact over $\mathbb{Q}$, 2 points |
| 25 | $625 \times 625$ | PD, $1.45 \times 10^{-7}$ | PD, $5.80 \times 10^{-9}$ | $\mathbb{F}_p$, 3 primes, 2 points |

Table 4: Independent verification of the uniform certificate. Entries under $LDL^T$ are the smallest exact pivot.

Because the definiteness check runs on the unblocked matrices, it is an independent confirmation of the block theory of §6.2 as well as of the numbers.

**Why $\mathbb{F}_p$ at $n = 25$, and why that is still exact.** The rational evaluation would need about $2.4 \times 10^8$ `Fraction` multiplications per point. The same computation modulo a prime is exact arithmetic — not floating point — and vectorises: with $p < 2^{20}$ every intermediate stays inside 64-bit integers, so the 625 quadratic forms become 625 matrix–vector products. Run at three unrelated primes near $2^{20}$, a surviving discrepancy would have to be divisible by all three.

**Positive control.** The same verifier is run on the independently produced, already-verified **stored** certificates at $n = 4$ and $n = 5$, using their own orbit data rather than any mapping of ours, and accepts them. If that had failed the verifier would have been the thing at fault.

**Negative controls, four of them.** Three single-variable perturbations of the uniform certificate at $n = 4$ — variables 0, 5 and 16, each moved by $10^{-6}$ — are rejected over $\mathbb{Q}$; a perturbation at $n = 6$ is rejected in $\mathbb{F}_p$ at all three primes. A verifier that never rejects proves nothing.

## 7.1 The fixed-dimension certificates

**Support layer: exact rational certificates, accepted by a standalone verifier. These are not Lean-checked, and no theorem of this paper depends on them.**

Five certificates at fixed dimensions were produced independently of the uniform construction, each with its own numerical solve and its own rounding. Each establishes the stated bound on the stated $K_n$, with equality only at $J_n/n$: $\Phi_3 \le 16/9$ on $K_3$; $\Phi_3 \le 61/32$ on $K_4$; $\Phi_3 \le 244/125$ on $K_5$; $\Phi_3 \le 71/36$ on $K_6$; and, at a second value of $k$ in the intermediate range, $\Phi_4 \le 1226/625$ on $K_5$.

The four $k = 3$ cases are covered by Theorem A and Corollary B, equality cases included. They are retained because they are the anchors against which the uniform certificate was checked, and because they were rounded independently of it. The case $(5, 4)$ is not reached by the uniform construction.

| $(n, k)$ | **bound** | **basis deg** | **monomials in $F$** | **cone** | **certificate** |
|---|---|---|---|---|---|
| $(3, 3)$ | 16/9 | 2 | 93 | 54 | 314 rationals |
| $(4, 3)$ | 61/32 | 1 | 552 | 16 | 19 rationals |
| $(5, 3)$ | 244/125 | 1 | 2225 | 25 | 19 rationals |
| $(6, 3)$ | 71/36 | 1 | 6906 | 36 | 19 rationals |
| $(5, 4)$ | 1226/625 | 2 | 7875 | 350 | 440 rationals |

Table 5: The five fixed-dimension certificates. Every one is exact rational data; the identity is confirmed by **full coefficient comparison**, not by sampling.

Smallest exact $LDL^T$ pivots, as the standalone verifier reports them — first the $\sigma_0$ Gram, then the smallest over **all** $n^2$ multiplier Grams: $8.24 \times 10^{-3}$ and $8.36 \times 10^{-3}$ at $(3, 3)$; $6.94 \times 10^{-3}$ and $6.65 \times 10^{-3}$ at $(4, 3)$; $2.02 \times 10^{-3}$ and $1.93 \times 10^{-3}$ at $(5, 3)$; $2.72 \times 10^{-4}$ and $6.48 \times 10^{-4}$ at $(6, 3)$; $5.83 \times 10^{-4}$ and $5.87 \times 10^{-4}$ at $(5, 4)$. Mutation tests perturbing a single coefficient by $10^{-20}$ are rejected in every case.

The second number needs its matrix named, because two are in circulation. The $n^2$ multiplier Grams are permutation conjugates of one another and so share a spectrum, but an $LDL^T$ pivot depends on the order of the indices, and the smallest pivot is not constant across the orbit. The design pipeline factors the Gram at the corner $p = (0, 0)$ and reports that one — $7.32 \times 10^{-3}$ at $(4, 3)$, $7.11 \times 10^{-4}$ at $(6, 3)$ — where the verifier factors all $n^2$ and reports the minimum. Both decide the same question, and the verifier's figure is the conservative one.

All five cases pass the standalone verifier in full — six checks each, standard library only, no shared code with the pipeline. In case $(5, 4)$ the exported data writes all 26 Gram matrices out densely so that the verifier needs no group theory, so it runs 26 exact rational factorisations of size 350 where only two are distinct. That redundancy is deliberate: exploiting the conjugacy would mean either trusting the conjugating permutation or verifying it, which is exactly the group theory the dense export exists to keep out of the verifier. No matrix is skipped.

$(5, 4)$ also needed the block-diagonalisation to be solvable at all: at cone size 350 a monolithic interior-point solve needs of order 30 GB per cone. Blocked, $\sigma_0$ gives cones $[5, 5, 4, 2, 2, 1, 1, 1, 1, 1]$ and $\sigma_{11}$

gives $[16, 14, 10, 7, 4, 4, 3, 2, 1, 1, 1]$ — identical to the blocks already recorded for the Dittert case at $n = 5$ — with largest cone 16.

# 8 Confinement, for every $k$

Theorem E is recalled from [6] and transferred to all $k$; nothing in this section is claimed as new mathematics. It is included because it is uniform in $k$ where the certificate of §5 is not, and because it is short enough to formalise once for the whole family.

The proof needs no expansion of the objective. Split the deficit as

$$(2 - \gamma(n, k)) - \Phi_k(A) = [1 - E_k(r)] + [1 - E_k(c)] + [P_k(A) - \gamma(n, k)], \tag{20}$$

where the third bracket is bounded below by $-\gamma(n, k)$ because $P_k$ of a non-negative matrix is non-negative. For the first two brackets, Maclaurin's inequality in the telescoped form of Theorem F gives $E_k(r) \leq E_2(r)$ on the simplex, and $E_2$ is exactly computable:

$$E_2(r) = 1 - \frac{\sum_i (r_i - 1)^2}{n(n-1)} \quad \text{whenever} \sum_i r_i = n, \tag{21}$$

which is `SubDittertK2.E_two_eq` and needs no positivity. Together these give Theorem E, and its contrapositive is the confinement statement: a violator of the Cheon–Hwang bound has its line sums within $(n-1)k!/n^{k-1}$ of $\mathbf{1}$ in squared $\ell^2$ distance, a neighbourhood that shrinks in $n$ for every fixed $k \geq 2$.

The hypothesis $E_k \leq E_2$ is stated in Lean as a named predicate, `SubDittertM.MaclaurinBound`, and `SubDittertMaclaurin.maclaurinBound_holds` discharges it for every $2 \leq k \leq n$; `theoremM'` and `confinement'` are the resulting unconditional statements. At $k = 2$ the predicate holds with equality, so the conclusion is not vacuous there.

## 8.1 Newton's and Maclaurin's inequalities in Lean

Theorem F is classical, and the contribution is the formalisation. The route avoids the reversal of polynomials that the textbook proof uses and whose root theory Mathlib does not carry. Writing $Q_i = \mathrm{coeff}_i / \binom{m}{i}$ for a polynomial of degree $m$, one has $Q_i' = m \cdot Q_{i+1}$, so Newton's inequality at index $i$ for $p$ is Newton's inequality at index $i - 1$ for $p'$, and $p'$ is real-rooted whenever $p$ is. That reduces every index $i \geq 2$ to a lower degree. Differentiation shifts the index upward, so $i = 1$ is never reached by the induction; it is not a base case at one degree but has to be proved directly at every degree, and it is where real-rootedness is consumed. The ingredients are Rolle's theorem for real-rooted polynomials, Vieta's formulas, the reciprocal identity $e_{n-j}(v) = \left(\prod_i v_i\right) \cdot e_j(v^{-1})$, and Chebyshev's sum inequality.

A second, independent Lean proof of Theorem F exists, written without sight of the first and by a different route — Mathlib's `coeff_eq_esymm_roots_of_card` with an induction on the degree, against an induction on the index — and its statements agree with those above. It is stored as verification evidence outside the build, in its own namespace (`NewtonCrosscheck`); agreement between two independent proofs is evidence that the statements are the intended ones, which is the one thing a kernel cannot check.

# 9 Formalisation

## 9.1 The verification standard

The development elaborates with no `sorry`, and no declaration in it depends on `sorryAx`. `native_decide` is used nowhere. Every declaration this paper cites in support of a stated result carries a `#print axioms` command, and each returns exactly `[propext, Classical.choice, Quot.sound]` — with one exception, `NewtonIneq.choose_mul_sub`, a statement about binomial coefficients, which returns the strict subset `[propext, Quot.sound]`.

All counts here are of the Lean sources as committed at `32c811e`, the commit at which they were last changed. The eight files carrying the results of this paper contain 511 axiom-audited declarations: 377 in `SubDittertK3.lean` and 7 in `RookSum.lean`, which together prove Theorem A; 21 in `SubDittertUniversal.lean` (Theorem C), 17 in `SubDittertK2.lean` (Theorem D), 37 in `SubDittertLinear.lean`, 13 in `SubDittertM.lean` and 6 in `SubDittertMaclaurin.lean` (Theorem E), and 33 in `NewtonInequalities.lean` (Theorem F). In `SubDittertK3.lean` the 377 are **every** named declaration in the file, so nothing in the file carrying Theorem A is left unaudited. The independent cross-check of §8.1 adds 12 more and is deliberately outside the build.

In the other files the audits cover the results and their supporting lemmas rather than every private definition. That is not a gap: `#print axioms` reports the axioms of the **transitive closure** of a proof term, so auditing a theorem audits every definition and lemma the theorem rests on. No file contains a `sorry`, so nothing in any of them can acquire `sorryAx` from within; and an unaudited declaration on which no stated result depends cannot affect a stated result.

The development is built against Lean 4.14.0 and Mathlib `v4.14.0` [13]. `NewtonInequalities.lean` imports Mathlib and nothing else — no permanent, no $K_n$ — because the results it proves are library material and the boundary should stay visible.

What the kernel cannot check is the translation of the 1992 statement into Lean's definitions. That is what §9.2 item 2 defends, and it is the only place left where a human error would survive the machine.

## 9.2 Theorem A, in the order it is built

**`subDittert_k3_full` is Theorem A, in all three parts.** For every $n \geq 4$ and every $A$ with non-negative entries summing to $n$, it proves $E_3(r) + E_3(c) - P_3(A) \leq 2 - 6/n^3$; that equality holds if and only if $A = J_n/n$; and the stability bound Equation 5 with the explicit $\theta_2(n)$ — with every definition in those statements built from scratch inside Lean.

1. **The statement.** `Kn`, `rowSum`, `colSum`, `esym`, `subPerm`, `sigmaK`, `E`, `P` and `Phi` are defined from scratch, with $\sigma_k$ a double sum over `Finset.powersetCard k univ` of the permanent of the submatrix on rows $S$ and columns $T$. The final statement `subDittert_k3` is written out in the 1992 notation.
2. **Validation of the definitions**, which matters more than it looks. `Phi_uniform` proves $\Phi_k(J_n/n) = 2 - k!/n^k$ for **every** $k \leq n$, so the functional as formalised is tight at the conjectured maximiser exactly where the conjecture says it is; instances at $n = 3, 4, 5$ specialise it to $2 - 6/n^3$. `sigmaK_rankOne` proves $\sigma_k(xy^T) = k! \cdot e_k(x) \cdot e_k(y)$, which is the check that rows are read from $S$ and columns from $T$: a definition that reads both indices off the same subset satisfies a different identity, and a symmetric test matrix cannot detect that error, whereas this does. A concrete $\sigma_3 = 450$ on an explicit $3 \times 3$ matrix is checked against a hand value.
3. **The combinatorial half, closed for all $n$ at once.** `RookSum.lean` proves a closed form for $\sigma_3$ directly from the subpermanent definition: the bridge is a bijection between $k$-subsets paired with a permutation of them and injections Fin $k \to$ Fin $n$, stated and proved general in $k$, specialised to $k = 3$ by a three-index inclusion–exclusion sieve. On top of that, `obj_eq_objPoly` proves that the objective $(2 - 6/n^3) - \Phi_3(A)$ equals an explicit polynomial `objPoly n A` in the entries of

$A$ and its row and column sums, for every $n \geq 3$. Together these eliminate `Matrix.permanent`, `subPerm`, `sigmaK`, `esym` and every sum over `Finset.powersetCard` from everything downstream: what remains of the proof mentions none of them. `objPoly` is cross-validated outside Lean against the from-the-1992-definition objective of §7, exactly over $\mathbb{Q}$, at random rational points for $n = 4, 5, 6, 7$, at $J_n/n$ where it must vanish, and against a mutation control.

4. **The ten positivity facts.** `certPositive_of_four_le` proves all ten quantities of §6.6 positive for every **real** $n \geq 4$ — real, not merely natural, which is strictly stronger and no harder here. Sturm is not needed inside Lean: after the substitution $n = m + 4$ all twenty polynomials involved (ten numerators, ten denominators) have non-negative coefficients and a positive constant term, so each proof is one `ring` and one `linarith`. As noted in §6.2, one of the ten — `H.D` — is not independent ($D = \theta_2/n$ exactly), so this proves nine independent facts and one immediate consequence of them; Lean proves all ten on the same uniform footing regardless, since the substitution argument costs nothing extra to repeat.
5. **Both Gram families, and the $\sigma_0$ Gram positive definite.** `G0_posSemidef` and `Hm_posSemidef` prove semidefiniteness for every $n \geq 4$, with no spectral theorem in either, and `G0_posDef` upgrades the first to **definiteness** — which is what the equality case needs. The $\sigma_0$ form splits into four manifestly non-negative pieces, one of them $\theta_2(n) \cdot \sum_u b_u^2$; since $\theta_2(n) > 0$ is already among the ten facts, that piece alone is strictly positive off the origin, so definiteness costs ten further lines and no new mathematics. The eigenvalue multiplicities of §6.2 are not needed for it. For a multiplier Gram the coordinates are the residuals about the corner's own row and column; in those the form breaks into blocks $A$, $B$, $C$, $D$ — the isotypic blocks of §6.2, in this basis rather than the one §6.2 normalises them in — and each is bounded below by a completion of squares, Lagrange's identity in two variables and Sylvester's in three. Seven positive quantities drive that, and they are the leading principal minors of those four blocks **times the constants** $1, 2, 4, 2, 1, 4, 1$. The constants do not depend on $n$, so positivity is the same question either way, but a bridge lemma stated without them is false. They are checked outside Lean as identities of rational functions, and the change of coordinates exactly over $\mathbb{Q}$ at $n = 4, 5, 6$ at three corners, with a mutation control.
6. **The Positivstellensatz identity.** `certificate_identity` proves Equation 14 itself, for every $n \geq 4$ and every $A \in K_n$, with $G_0$ and the family $H_p$ the explicit Grams above. Both sums over corners are expanded in ten global invariants of the centred coordinates. Every coefficient in that expansion was emitted rather than typed, and checked outside Lean as an identity of rational functions of $n$ by full coefficient comparison rather than by sampling.
7. **The deduction.** `subDittert_k3_of_certificate` derives the bound from the certificate through the bridge lemma: non-negativity of the quadratic forms from `PosSemidef`, non-negativity of the entries of $A$ from membership of $K_n$, then `linarith`. The $\lambda(b) \cdot \left(\sum_q b_q\right)$ term does not appear, because that sum vanishes on $K_n$.
8. **The equality case and the stability bound**, both for every $n \geq 4$. `subDittert_k3_bound_and_uniqueness` is the bridge lemma's packaged form `certificate_bound_and_uniqueness` applied to the certificate's own data; its separation hypothesis is `eq_uniform_of_centre_eq_zero`, which is the remark that the centred coordinates are $b_p = A_p - 1/n$ entrywise, so their vanishing **is** $A = J_n/n$. `subDittert_k3_stability` keeps the $\theta_2(n) \cdot \sum_u b_u^2$ piece instead of discarding it and so proves Equation 5, with `subDittert_k3_stability_explicit` the same statement with the constant written out; `eq_uniform_of_Phi_eq_of_stability` then re-derives uniqueness from stability alone, without the positive-definite bridge, so the equality case stands on two independent Lean proofs. `subDittert_k3_full` states all three parts of Theorem A as one theorem in the 1992 notation.

**A normalisation caveat, stated exactly**, on the ten quantities of the fourth item — a separate matter from the seven constants of the fifth. Each numerator and denominator in the Lean file is separately cleared to primitive integer coefficients, and the pair is sign-flipped together where the denominator is negative on $n \geq 4$. Both operations scale by a positive rational, so each Lean quantity is a constant **positive** rational multiple of the corresponding quantity in the Sturm pipeline; the multiple is 1 except for `H.A minor2` and `H.B` (both 1/2) and `H.C minor2` (4). **These four numbers are an artefact of two independent choices of primitive-integer clearing, one on each side, and carry no mathematical content of their own**; a different but equally valid clearing convention on either side would change 1/2 and 4 to other positive rationals without changing anything else. This was cross-checked at $n = 4, 5, 7, 13, 101$: the ratio is constant and positive in every case, so positivity of one is positivity of the other, which is the only fact that matters.

## 9.3 Theorems C to F in Lean

The uniform half of the development is independent of the certificate and much shorter.

`universal_identity` (Theorem C) is proved for every $n$ and every $k \leq n$ from two coefficient rules. The $e_k$ rule is the centre identity `esym_one_add` of `SubDittertLinear.lean`; the $\sigma_k$ rule, `sigmaK_one_add`, is proved through a rook bridge valid at every $k$ — summing over pairs of injections Fin $k \to$ Fin $n$ is $k!$ times the subpermanent sum — with the fibres of the restriction map counted by Mathlib's `Equiv.sumEmbeddingEquivSigmaEmbeddingRestricted`. One supporting lemma, `choose_subset_of_subset`, is absent from Mathlib and is a candidate for upstreaming.

`subDittert_k2` (Theorem D) instantiates `universal_identity` at $k = 2$, closes $\sigma_2$ and $e_2$ in closed form by the $k = 2$ sieve, and concludes through the same bridge lemma the $k = 3$ proof uses. There is no Gram matrix in it.

`theoremM'` and `confinement'` (Theorem E) are `theoremM` and `confinement` with the hypothesis `MaclaurinBound` discharged by `maclaurinBound_holds`. The attribution to [6] is the first section of the header of the file that states them.

`newtonAt_all`, `newton_esymF` and `pnorm_le_two` (Theorem F) are proved in a file that imports Mathlib alone.

## 9.4 Equality and stability

For each of the five fixed-dimension certificates of §7.1, equality holds **only** at $J_n/n$. The argument is one line: the $\sigma_0$ Gram is positive **definite** by exact $LDL^T$, and its monomial basis excludes the constant, so $\sigma_0(b) = 0$ forces $b = 0$, while every other term of Equation 14 is non-negative on $K_n$.

For every $n \geq 4$ the same conclusion is Lean-proved, and it needs no part of the block-diagonalisation of §6.2. The Bose–Mesner form of the $\sigma_0$ Gram makes its quadratic form

$$\sigma_0(b) = c_0^{\text{tot}} \left( \sum_u b_u \right)^2 + c_0^{\text{line}} \left( \sum_i R_i^2 + \sum_j C_j^2 \right) + \theta_2(n) \sum_u b_u^2 \tag{22}$$

identically in $b$, with all three coefficients positive on $n \geq 4$; the identity is `quadForm_G0`. The last term alone is strictly positive unless every $b_u$ vanishes, so definiteness follows from $\theta_2(n) > 0$, one of the ten facts of §6.6, by an argument that mentions neither eigenvalues nor their multiplicities. The spectral decomposition is the route by which the coefficients were found; the finished identity does not depend on that route.

Keeping the $\theta_2$ term of Equation 22 rather than discarding it costs nothing and gives Equation 5, which is strictly stronger than the equality case. The constant is **not** claimed optimal. Both sides of Equation 5 were evaluated exactly over $\mathbb{Q}$ at $n = 4, 5, 6$ and the slack ratio measured. Along directions where the row and column sums of $A - J_n/n$ all vanish, the first two terms of Equation 22 vanish identically, only the multiplier half of the certificate is being discarded, and the ratio is constant in the size of the perturbation — 2.88, 4.34, 5.70 at $n = 4, 5, 6$. So $\theta_2(n)$ is within a factor of about three to six of the best constant in the directions where it is closest to tight, and further off in general directions. Sharpening it would mean retaining part of the multiplier half, which we have not attempted.

### 9.5 What is not formalised

One thing is not: the block-diagonalisation of §6.2 as an **equivalence** — the Bose–Mesner eigenvalues with their multiplicities, the isotypic decomposition, and with them the statement that positive definiteness of the two Grams is exactly the positivity of the ten rational functions. Lean proves the implication the theorem needs and not the converse, and it proves it by explicit algebra rather than by that route, so the reduction of §6.2 is corroborated by the Lean development without being verified by it. The two are checked against each other outside Lean: the unblocked $n^2 \times n^2$ matrices are factored directly at six dimensions in §7.

The fixed-dimension certificates of §7.1 are also unformalised, as §1.2 records.

## 10 Computational evidence toward $k = 4$ and $k = 5$

**Support layer: exact rational computation with stored witnesses. Nothing in this section is a theorem of this paper, and nothing in it is Lean-checked.**

The certificate programme of §5 has a shape that depends on $k$ only through $e = \lceil (k-1)/2 \rceil$, because a common Gram basis of degree $e$ gives $\deg\big(b_p \sigma_p\big) = 2e + 1$, which must reach $\deg F = k$. Mixed degrees are ruled out for even $k$: if $\sigma_0$ carried basis degree $k/2$ and the multipliers $k/2 - 1$, the degree-$k$ part of the identity would make the top form $F_k$ a sum of squares on the hyperplane, and Equation 7 shows $F_k(b) = s_k[s_k \sigma_k(b) - e_k(R) - e_k(C)]$, which is negative at explicit doubly centred integer matrices — the smallest is a $4 \times 4$ with $\mathrm{per}(b) = -854$. So $e(k) = \lceil (k-1)/2 \rceil$ and the programme's counts are constant on the bands $\{2, 3\}$, $\{4, 5\}$, $\{6, 7\}$, and so on: within a band the constraint matrix is literally the same matrix and only the right-hand side moves. Band one is $\{2, 3\}$, which Theorems A and D close.

The same computation bounds the method's reach. The binding block's multiplicity is $2, 16, 93$ at $e = 1, 2, 3$, and the programme grows with it: 19 unknowns at $e = 1$ and 440 at $e = 2$. The route therefore reaches every **fixed** $k$ at all large $n$ and cannot reach $k = n$, where the stabilisation threshold and the parameter collide.

For band two the current state is a decided sweep rather than a certificate. At $k = 4$ the pinning programme was decided over $\mathbb{Q}$ at $n = 5, 6, 7$, with no verdict resting on a solver's reported sign: infeasibility is witnessed by a rational Farkas multiplier, positive semidefinite by construction, whose value is constant and non-positive on the affine set; feasibility by a rational point with every canonical block positive definite under exact $LDL^T$. The full pinning is infeasible at all three dimensions, as are twelve of the thirteen single-block omissions. The one programme that survives — 201 pins, omitting the $16 \times 16$ block of $\sigma_{11}$ — has an exact strictly feasible rational point at $n = 5$, with least pivot $5.85 \times 10^{-5}$, and is boundary-undecided at $n = 6$ and $n = 7$. Every verdict has its witness stored as exact rationals and re-checked by a standalone verifier that re-derives the constraint system and the pin rows from the problem definition, carries its own elimination and its own $LDL^T$, and confirms that every canonical basis has full column rank — without which block definiteness would not imply definiteness of the whole and the test would be vacuous. Mutation controls were run and failed as they should.

There is no certificate family in $n$ for band two, so there is nothing yet to formalise.

## 11 Assessment

**What is proved.** Theorem A, for every $n \geq 4$, by an exact rational certificate whose coefficients are rational functions of $n$, with definiteness decided by Sturm sequences over $\mathbb{Q}$ and the identity verified independently at six dimensions. Theorems C to F, in Lean, uniformly in $n$ and $k$. No floating-point number enters any of these decisions: the numerics find candidates, and every accepted statement is re-established over $\mathbb{Q}$ or over $\mathbb{F}_p$.

**What kind of proof this is.** Theorem A is a computer proof, and a small one by the standards of the genre — nineteen rational functions of $n$, of degree at most 12. It is checkable in minutes at any fixed $n$. It conveys very little insight into **why** the bound holds; §6.3 is the closest thing to an explanation we can offer, and it is an explanation of the certificate's geometry rather than of the inequality. Theorems C to F are ordinary mathematics that happens to be machine-checked.

**What is machine-checked, precisely.** Theorem A itself in all three parts, and with it every step between the 1992 statement and them: the definitions, the tightness of the functional at $J_n/n$ for every $k \leq n$, the rank-one identity that pins the definition of $\sigma_k$, the closed form for $\sigma_3$ and its reduction of the objective to an explicit polynomial, the ten positivity facts for all real $n \geq 4$, positive semidefiniteness of both Gram families and positive **definiteness** of the $\sigma_0$ Gram, the Positivstellensatz identity, the deduction from certificate to bound, the equality case — twice, by independent routes — and the stability bound with its explicit constant. Beyond Theorem A: the universal decomposition at every $(n, k)$, the case $k = 2$ at every $n$, the confinement at every $2 \leq k \leq n$, and Newton's and Maclaurin's inequalities. §9.1 states the standard and pins it to a commit. What the machine does not check is that the Lean definitions say what the 1992 paper says; §9.2 item 2 is the argument that they do. We state the division explicitly because "machine-checked" is otherwise an ambiguous claim.

**What is not claimed.** No case with $k \geq 4$ beyond the fixed-dimension certificate at $(5, 4)$, which is not Lean-checked; no statement about general $k$ beyond Theorems C and E; nothing refereed. The constant $\theta_2(n)$ of Equation 5 is not claimed optimal (§9.4). No novelty is claimed for the identity of Theorem C, for the statement of Theorem D, for Theorem E, or for Theorem F. We have not read Cheon and Wanless 2007 [7] and do not rely on it.

**On the priority claim.** §2 records public dated claims on the $k = n$ endpoint that appeared within a single week of this work, three of them in public repositories [11] with no corresponding preprint. To our knowledge Theorem A is the first resolved case of the Cheon–Hwang conjecture with $2 < k < n$; the claim covers public repositories as well as the indexed literature, and it is a priority claim, not a claim of difficulty. Such a claim is perishable on a line that has moved this fast, and it should be re-checked before this paper is submitted anywhere.

**Where the method stops.** $\deg F = k$, so the Gram basis degree must grow with $k$, and the degree-counting obstruction that blocks the $k = n$ line at even dimensions reappears in $k$; §10 quantifies it. The line $k = 3$ is tractable exactly because its degree budget is fixed while the dimension grows. The natural next targets are $(6, 4)$ and $(6, 5)$ by the fixed-dimension method, and — a genuinely different question — whether $k = 4$ admits the same uniform treatment as $k = 3$.

## 12 Data availability

The certificates as exact rational data, the objective builders, the block-diagonalisation, the Sturm decision, the independent verifiers with their positive and negative controls, and the Lean development are supplied as a reproduction kit accompanying this paper. Its README carries the file manifest, naming the claim each file backs and the procedure for re-running each check. The failed design branches are retained there, because §6.5 is only checkable against them.

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[5] G.-S. Cheon and I. M. Wanless, *An update on Minc's survey of open problems involving permanents*, Linear Algebra and its Applications **403** (2005), 314–342.

[6] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra and its Applications **436** (2012), 791–801, `doi:10.1016/j.laa.2010.08.041`. (Theorem 2.1 is Theorem E at $k = n$.)

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[9] Z. Pang, arXiv:2606.01531 (1 June 2026). Dittert for $n \geq 17$; preprint, no DOI, not refereed.

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[11] Public repositories claiming Dittert cases, all created 21–24 July 2026, none refereed, all accessed 28 July 2026: `123ljh0bot/Dittert_Conjecture_in_Dimension_4` ($n = 4$, Lean 4 with an SOS certificate); `pedromnasc/dittert-conjecture-proof` ($n = 4, 5, 8$–$16$, and the `subdittert/` package quoted in §2.2); `lueluelue2006/dittert-conjecture-draft` and `lueluelue2006/dittert-n7-extension`, attributed to Hongyuan Lu ($n = 6$–$15$). Each is at `https://github.com/` followed by the owner and name shown.

[12] G.-S. Cheon and S. Yoon, *A note on the Dittert conjecture for permanents* (2006); G.-S. Cheon, *On the monotonicity of the Dittert function* (1993). The remaining two of the five papers citing [1]; neither works on the generalisation.

[13] The mathlib Community, *The Lean mathematical library*, CPP 2020.